Collision Course

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1 Abstract

Dark matter halos grow through hierarchical mergers, but the way these interactions reshape the internal structures remains difficult to predict. In this project, I simulate controlled collisions between two isolated dark matter halos using a collisionless N-body code with Barnes–Hut force calculations. To compare major and minors mergers, the mass ratio between the halos was varied and the systems were evolved until the remnant reached a stable virialization. The concentration of the larger 'host' halo was calculated by fitting the density profile to an NFW model over time. As expected, all merger types alter the halo concentration by a sudden decrease after merging followed by steady increase as the system relaxes. However, mid range mergers bounced between minor and major merger concentration values suggesting that the relation between merger mass ratio and the changes in internal density are not linear.

2 Introduction

Throughout the twentieth century, astronomers have observed that the gravitational dynamics of spiral galaxies could not be described by their visible (baryonic) mass alone (Rubin 1983). This 'missing' mass problem led to the introduction of cold dark matter, which constitutes the majority of the universe, but only interacts gravitationally (Blumenthal and Faber 1984). The current standard for describing cosmic interactions is the Lambda Cold Dark Matter (ΛCDM) cosmological model, which combines the cold dark matter theory with the cosmological constant (Λ) representing dark energy. This model currently produces simulations that are closest to modeling the statistics and behaviors of deep space observations (Klypin et al. 2011).

According to the ΛCDM model, dark matter masses grow hierarchically: smaller clumps merge until they form large spherical structures called dark matter halos (Wang et al. 2020). These halos have large potential wells that accrete additional material such as gas and interstellar dust. Over time, the centers of the halos may become dense enough that the baryonic (visible) matter collapses and forms a galaxy at the center of the halo (Blumenthal and Faber

1984). Therefore to predict galaxy formation, the central density of a halo must be tracked through time.

The central density of a dark matter halo is described by its concentration ($c_{\rm vir}$) which is a measure of how rapidly the density grows toward the center. If a halo has a higher concentration, its baryonic matter is more likely to collapse into a galaxy. The biggest changes to a halo's concentration occur during merger events because they disrupt the potential field of the halo, shaking up the particle positions. Since there is an abundance of small halos compared to large galaxy hosts, most merger events occur between a large 'host' halo and a much smaller satellite halo (Klypin et al. 2011).

As a satellite halo falls into a large halo's potential well, its orbital energy is transferred to the host primarily through two gravitational processes: dynamical friction and tidal shocks. Dynamical friction is the drag force exerted on a satellite by the host's particles. The satellite's gravitational acceleration increases the kinetic energy and velocity of the host halo's particles (a gravitational wake), causing them to spread out, and decreasing the density in the region in which the satellite is passing (Ma and Boylan-Kolchin 2004). Because the deceleration force is proportional to the background density of the host halo, the effect increases as the satellite moves to smaller radii.

Tidal shocks occur when the host halo's potential field changes rapidly compared to the orbital period of its particles, like in a major merger. Each halo exerts significant tidal forces on the other's loosely bound outer particles, injecting energy and accreting material. As their center of masses approach each other, the particles start to move between the two systems and the potential field becomes highly time dependent (Valluri et al. 2006). As each halo's particles increase in velocity away from their bound system, their internal density will decrease until the original potential wells combine into a single remnant.

These two processes are expected to disperse the dark matter particles in the host halo during mergers theoretically, but they have also been observed in large scale ΛCDM simulations like the Millenium Simulation (Springel et al. 2005) or the FIRE simulations (Hopkins et al. 2018). These simulations show merger driven evolution that reshapes halo structure and effects star formation, but they do not have a clear relation between the mass ratio of the merger and the changes in central density.

3 Literature Review

There are several smaller studies that have begun investigated the effects of halo mergers on their internal structure and their evolution over time. Wang et al. (2020) found that in CDM simulations, the halo concentration experiences a sharp decrease following a major merger and then a gradual increase to a slightly higher concentration than before. More importantly, they found that minor mergers caused similar, but weaker changes in the internal density directly following the collision. They found that the cumulative effects of major and minor mergers lead to scattering in concentrations at fixed halo mass (and

formation time). This suggests that the relation between the mass ratio and concentration may not be straightforward.

Other studies looked at the dynamics of long-term merges between large halos and orbiting satellite halos: Kazantzidis et al. (2006) found that halo satellites orbiting a larger halo do not have a significant mass redistribution despite being tidally stripped. Therefore the density distribution is relatively stable, so the satellite halo's concentration is constant over time as well.

Although Kazantzidis et al. did not find significant changes in the density profile of the satellite halos, there was a study done by Ma and Boylan-Kolchin (2004) on the changes in the host halo's density profile. They simulated a CDM halo with 1000 satellites and analyzed the resulting density profile compared to a halo with no satellites. When present, the inner density of the host halo flattened out to a core-like inner curve. This causes the concentration to decrease as well, consistent with the findings of Wang et al. (2020).

4 Problem Statement

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The assembly history of a dark matter halo determines the evolution of its internal structure and is therefore critical in modeling galaxy formation in the ΛCDM model. Previously mentioned studies like Wang et al. (2020), found that all size mergers affect the internal structure, but the relationship between the mass ratio and the change in concentration is not easily defined with a linear relation. In particular, the relative impacts between major and minor mergers on the final density profile remain uncertain despite the rate of minor mergers far out-weighing the rate of major mergers (Klypin et al. 2011). This highlights that tracking halo concentration requires detailed merger histories and a thorough understanding of the dynamics between mergers and internal halo structure.

The goal of this project is to determine how mergers of different mass ratios affect the internal structure of a halo, specifically the concentration, and to compare the effects between major and minor mergers.

To investigate this behavior, we simulated the controlled collision of two dark matter halos in a dark matter only environment. Each dark matter halo will be modeled as a system of collisionless particles of equal mass so that we can analyze their internal dynamics throughout the merging process. Baryonic matter like gas and stars will not be included because baryonic effects lie beyond the scope of this study: they would complicate our interpretations of the dark matter dynamics. Dark energy and relativistic effects will also be neglected, because the velocities of objects in our simulation do not approach the speed of light.

Newtonian gravity is the only force in the simulation, so the isolated halos will exert a large gravitational force on each other, and will ultimately collide and merge to become one 'remnant' halo. The net force on a single particle is equal to the sum of gravitational force from every other particle in the system,

so the equation of motion for a single particle is:

$$a_i = \sum_{i \neq j}^{N} \frac{G m_i \vec{r_{ij}}}{|\vec{r_{ij}}|^3} \tag{1}$$

where a_i is the acceleration of the ith particle, m_i is the mass of a single particle, G is the gravitational constant, and $\vec{r_{ij}} = \vec{r_i} - \vec{r_j}$ is vector between particle i and particle j whose force is being calculated. To find the concentration of the remnant halo, the system was evolved until the remnant had stable virialization, defined by the virial theorem:

$$T = \frac{U}{2} \tag{2}$$

Where T is the total kinetic energy of the system and U is the total potential energy of the system. The virialization is considered stable once the rate of change approaches zero. A stable virialization is required to calculate an accurate final virial concentration, which is the ratio of the virial radius to the scale radius of the halo:

$$c_{\text{vir}} = \frac{r_{\text{vir}}}{r_{\text{scale}}} \tag{3}$$

The scale radius of a halo is defined by the radius at which the density profile of the halo has a logarithmic slope of -2, thus a measure of where there is an increase in central density in a halo. The standard dark matter only density profile is the NFW profile (Navarro, Frenk and White 1996):

$$\rho(r) = \frac{\rho_{\text{scale}}}{(\frac{r}{r_{\text{scale}}})(\frac{r}{r_{\text{scale}}} + 1)^2} \tag{4}$$

where ρ is the density at radius r and ρ_{scale} is the density at the scale radius r_{scale}. To calculate the scale radius, the density profile of the remnant will be fitted to the NFW profile until a goodness of fit parameter Q defined in Lazar et al. (2020) is minimized (Eq. 5). The best fit parameter for scale radius will be used in Eq. 3 to calculate the final c_{vir}.

$$Q^2 = \frac{1}{N_{\text{bins}}} \sum_{i}^{N\text{bins}} [log_{10}\rho_i - log_{10}\rho_i^{model}]^2 \tag{5}$$

5 Methods

5.1 Halo Initialization

Each dark matter halo will be modeled as a system of collisionless particles, with fixed mass m. If all particles have the same mass, the total mass of each halo, M_i, is equal to the number of particles within it, N_i multiplied by the particle mass. To simulate different merger scenarios, the satellite halo will remain fixed at the minimum particle count required for a physically reliable density profile

and gradually increase the number of particles in the host halo, varying the mass ratio μ.

Thus, varying the mass ratio μ is equivalent to varying the ratio of the number of particles in each halo where M_{host} and $M_{satellite}$ are the total mass of each halo and N_{host} and $N_{satellite}$ are the total number of particles in each halo.

$$\mu = \frac{M_{\text{host}}}{M_{\text{satellite}}} = \frac{N_{\text{host}}}{N_{\text{satellite}}} \tag{6}$$

The minimum number of particles required to accurately model the density and velocity distributions of a dark matter halo is $N_{\text{min}} = 4000$ particles (Poveda-Ruiz et al. 2016). This corresponds to a minimum halo mass of $M_{\text{min}} = 4 * 10^3 M_{\odot}$. Simulations have shown that the maximum halo mass for a halo without baryons is $M_{max} = 1 * 10^6 M_{\odot}$ (Benitez-Llambay and Frenk), which corresponds to $N_{\text{max}} = 100 N_{\text{min}} = 400,000$ particles. Thus, μ will be varied between 1 and 100.

Dark matter halos are spherically symmetric and isotropic, so the initial positions are defined by the radial density profile of dark matter only halos: the NFW profile (Navarro, Frenk, and White 1997). The parameter that defines the NFW profile is the scale radius. The scale radius is a characteristic radius in a halo profile that defines the transition between the diffuse outer layers of the halo and the denser inner region (Lazar et al. 2020). Assuming the halo is virialized initially, the scale can be found by solving Eq. 3 for r_{scale}.

The virial radius r_{vir} is given in terms of the total mass of the halo M_{vir}, the gravitational constant G, and the present day Hubble constant H_0 (Klypin et al. 2011).

$$r_{\text{vir}} = \left(\frac{G M_{\text{vir}}}{100 H_0^2} \right)^{1/3} \tag{7}$$

The initial concentration of each halo will always be $c_{\text{vir,i}} = 10$ which is an average value for smaller dark matter halos (Weschler et al. 2002), then r_{scale} can be calculated and used as a parameter in the NFW density profile (Eq. 4).

To sample initial positions from the mass density profile, 1000 logarithmically spaced shells will be created and the fraction of the total mass in each shell will be calculated using the integral:

$$M_{\text{shell}} = 4\pi \int_{r_1}^{r_2} \rho(r) r^2 dr = m N_{\text{shell}} \tag{8}$$

Each shell will be populated with N_{shell} particles with radial positions evenly distributed among the shell's radius range. Since the halo should be spherically symmetric, the particles will be assigned a uniform distribution of angular positions θ and ϕ. The centers of mass will be calculated and the halos will be 'placed' in the simulation box such that their centers of mass lie on the x axis, with the host at x = $r_{vir,host}$ and the satellite at the opposite value x = $-r_{vir,host}$.

The initial particle velocities will be given by the initial circular velocity distribution defined by Navarro, Frenk, and White 1997 (Eq. 9) where the

velocity of each particle is tangential to its radial position vector. The circular velocity of a particle at radius r (V_c), is given in terms of the gravitational constant G and the total mass contained within radius r: $M(< r)$.

$$V_c(r) = \sqrt{\frac{GM(< r)}{r}} \tag{9}$$

Both halos will have constant angular momentum in the z direction, so the circular velocity will be converted into initial cartesian velocities using the following equation:

$$\vec{V} = \frac{V_c * (\hat{z} \times \vec{r})}{|\vec{r}|} \tag{10}$$

5.2 Particle Evolution

Each halo system will be evolved using Newtonian physics, where the net force on each particle $\vec{F_{\text{net,i}}}$ is equal to the sum of the gravitational force of every other particle in the simulation. The gravitational force on the 'target' particle is given by Newton's gravitational law (Eq. 11) where G is the gravitational constant, m is the particle mass, $\vec{a_i}$ is the acceleration on the target particle, and $\vec{r_{ij}} = \vec{r_i} - \vec{r_j}$ is the displacement vector between the target particle and all other particles.

$$\vec{F}_{\text{net,i}} = m\vec{a}_i = \sum_{i \neq j}^{N} \frac{Gm^2(\vec{r_i} - \vec{r_j})}{|\vec{r_i} - \vec{r_j}|^3} \tag{11}$$

$$\vec{a}_i = \sum_{i \neq j}^{N} \frac{Gm\vec{r}_{ij}}{|\vec{r}_{ij}|^3} \tag{12}$$

The system is evolved using a leapfrog integration scheme, which is simple but effective for large N over many small fixed time steps (Quinn et al. 1997). This matches our scenario because all particles will be close enough that adaptive step sizes will not effectively reduce computational costs. Additionally, Quinn et al. 1997 found that the simple leap frog showed more accuracy and energy conservation over long periods of time than other methods like Runge Kutta, specifically in dark matter only systems.

The leapfrog integration works by staggering the updates to position and velocity: velocity is updated at half-step intervals, and position is updated at full-step intervals. The velocity and position update equations are given below, where i indicates the ith particle, t is the current timestep, h is the fixed step size, and a_i is the acceleration of particle i from Eq. 12. First, the velocity is advanced by a half step using the acceleration of the current configuration. That velocity is used to calculate the positions at the next full timestep, which are used to compute new acceleration and update the velocity a second time using the new configuration. This results in a second order accuracy approximation because it utilizes the midpoint velocity, while only requiring one force calculation per

step. This is preferred in large N-body simulations because the acceleration calculation is the dominant computational cost in the simulation.

$$\vec{v}_i(t+\frac{h}{2}) = \vec{v}_i + \frac{h}{2}\vec{a}_i \tag{13}$$

$$\vec{r}_i(t+h) = \vec{r}_i + h\vec{v}_i(t+\frac{h}{2}) \tag{14}$$

$$\vec{v}_i(t+h) = \vec{v}_i(t+\frac{h}{2}) + \frac{h}{2}\vec{a}_i(t+h) \tag{15}$$

The actual cost of computing the gravitational force *of* every particle *on* every particle using Eq. 12 scales with $\mathcal{O}(N^2)$ because there are N calculations on N particles. Our halos require a minimum of 40^3 particles, so the cost becomes increasingly expensive as we increase the mass ratio, μ. To reduce this cost, we will implement the Barnes–Hut hierarchical octree algorithm detailed by Barnes and Hut 1986. This method involves dividing the simulation space into octants: if there is more than one particle in an octant, it is recursively divided into octants until each suboctant or 'leaf' contains a unique particle. This creates a 'tree' in which each leaf stores the total mass of all particles inside and the location of the center of mass.

This framework allows us to shorten the force calculation by approximating the collective force from far away particles as the force from their octant's center of mass. The criterion for this approximation is that the side length of the octant L must be larger than the distance between its center of mass and the particle whose acceleration is being measured D (Barnes and Hut 1986). The parameter that represents this is θ, which typically has a value between [0.8 - 1]: if Eq. 16 is satisfied, then the force from that cell is computed as a point mass at the center of mass with mass equal to the total mass.

$$\theta > \frac{L}{D} \tag{16}$$

This reduces the computational cost of the leapfrog method from scaling as $\mathcal{O}(N^2)$ to $\mathcal{O}(NlogN)$ where it costs $\mathcal{O}(logN)$ to calculate the octree forces for a particular particle, and this must be done for all N particles.

The simulation will continue to evolve over time until it is virialization is stable, meaning the virial energy E_{vir} converges to zero: T is the total kinetic energy of the halo and U is the total potential energy of the halo

$$E_{vir} = T - \frac{U}{2} \tag{17}$$

Once this requirement has been met, the unbound particles that are no longer orbiting the halo are filtered out by finding the kinetic and potential energies of each individual particle:

$$T_i = \sum_{i=1}^{N} \frac{1}{2} m_{part} v_i^2 \tag{18}$$

$$U_i = -\sum_{j \neq i} \frac{G m_{part}^2}{|\vec{r_{ij}}|} \tag{19}$$

T and U are the sum of the kinetic and potential energies, respectively, of every particle in the simulation. m is the mass of a particle, $\vec{v_i}$ is the velocity of particle i, and $\vec{r_{ij}}$ is the distance vector between particle i and particle j, and G is the gravitational constant. To verify physical accuracy throughout the simulation, the total energy of the entire system will be calculated at all time steps and plotted versus time at the end of the simulation.

5.3 Concentration

The concentration of the remnant was measured every Gyr between the time that the halo merge and the time at which the system's virialization energy is stabilized. Stabilization was determined when the virialization energy had no large spikes for at least two Gyrs, across all merger types because they all end at same time. Before any parameters are found, unbound particles whose kinetic energy was greater than their potential were excluded from calculations. This is because they will escape their orbit of the center of mass and no longer belong to the halo.

Concentration is calculated using the virial radius and scale radius in Eq. 3. The virial radius is the maximum radius of the bounded particles from the center of mass, so it can be found from the particle positions directly. However, the scale radius can only be found by fitting the data to the NFW profile and determining the best fit parameters (one being r_{scale}).

To obtain the radial density profile of the data, we follow the reverse process from generating radial positions at the beginning. We assume that the remnant is spherically symmetric, so the data will be divided into 1000 spherical shells similar to before. The ratio of the number of particles in each shell to the shell's volume will give the mass density of each shell in $M_\odot/cm^3$. The corresponding radii array will be generated by finding the median radius of the particles in each shell. This profile will be fit to the NFW function (Eq. 4) using the function curve-fit from scipy.optimze in logarithmic space, and it will be repeated multiple times, tightening the bounds for the scale radius each time. The best fit will be determined using the minimum goodness of fit value given by Eq. 5 from Lazar et al. (2020). The scale radius parameter of that fit will be used to calculate the final c_{vir} using Eq. 3.

6 Results

We performed three controlled N-body simulations of dark matter halo mergers with each representing a class of merger: major, mid, and minor. The Two classes that are defined clearly in the literature are major and minor, but the mid class was added a 'control' group to rule out weird behavior and get a

full picture of the relation between mass ratio and change in concentration. The concentration was measured every Gyr between the time where the halos merge (around t = 2 Gyr) and when the virialization begins to stabilize in every simulation (around t = 6 Gyr) as shown in figure 1:

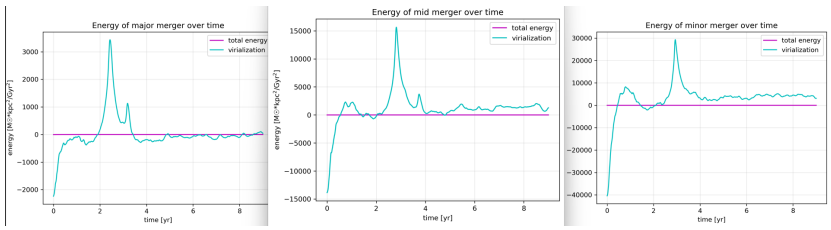

Figure 1: Total energy and virialization over time for each simulation: total energy is purple and virialization is light blue. From left to right: major merger, mid merger, minor merger.

The total energy was computed at every timestep in every simulation to show that energy is conserved throughout and the physical processes are accurate. In at least the major and mid mergers, the concentration of the halo decreased after the merge occurred then increased dramatically past the initial concentration of the host. This is shown in figure 2 where each colored line represents a mass ratio: major is the lightest color and minor is the darkest color. Although hard to see, the minor merger also decreased the concentration between t = 1 and t = 2 from the initial value cvir0 = 10 to cvir = 9.6 then increases concentration until the end of the simulation:

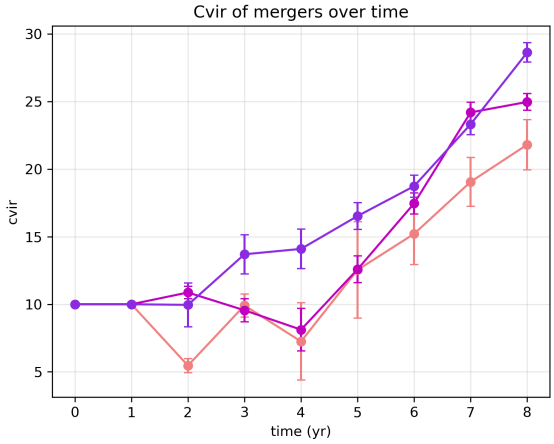

Figure 2: Cvir over time for each merger class: orange is major, pink is mid, and purple is minor. The black error bars are the goodness of fit Q values given by equation Eq. 5.

Most importantly, the major merger and minor merger concentrations followed nearly parallel growth patterns with the major merger's cvir being smaller, as expected from its degree of disruption. The mid merger, however, did not remain between the other two; it crossed over the minor merger cvir directly after the merge and then hopped up to the major's cvir after virialization stabilized. This behavior suggests that either the fitted concentrations in this simulation had higher uncertainty than that shown by the Q values or the relation between the mass ratio and concentration changes is intrinsically nonlinear. In either case, it highlights the need for further investigation of the relation between mergers and internal halo dynamics.

7 Discussion

Our results confirm that mergers of all sizes change the concentration of dark matter halos. The major merger produced the largest perturbations, but the mid merger produced effects that could not be explained linearly. This is the exact behavior found by Wang et al. (2020) in terms of both major and minor mergers.

This finding is significant for two reasons. First, small satellite halos have a lasting effect on the central density so the structure is sensitive to minor interactions. Second, the high abundance of low-mass intergalactic objects (like small satellite halos) in the universe (Klypin et al. 2011) implies that halos whose assembly histories dominated by minor mergers, like those in dense cosmic neighborhoods, will exhibit more scatter in concentration at a given mass.

These results highlight that halo internal dynamics are constantly and rapidly evolving. Consequently, accurately predicting galaxy formation within dark matter halos requires merger histories that track interactions down to very small mass ratios. While dark matter–only simulations are limited to modeling the evolution of dark matter itself, incorporating baryonic physics would allow a direct study of galaxy formation processes, as stars could form and gas dynamics could be captured alongside the dark matter evolution.

8 Bibliography

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